「Def. A pointwise linear map

$$\varphi : \bigcup_{p \in \mathbb{R}} T_p\mathbb{R}^3 \to \mathbb{R}$$

is called a 1-form on $\mathbb{R}^3$. The pointwise linear means; for fixed $p \in \mathbb{R}^3$,

$$\varphi_p : T_p\mathbb{R}^3 \to \mathbb{R} ; V_p \mapsto \varphi(V_p)$$

is linear functional of $T_p\mathbb{R}^3$.

Define the sum of 1-forms φ_1 and φ_2 as

$$(\varphi_1 + \varphi_2)(V_p) \overset{\text{def}}{=} \varphi_1(V_p) + \varphi_2(V_p) \text{ for any tangent vector } V_p.$$

And $f : \mathbb{R}^3 \to \mathbb{R}$ and 1-form φ,

$$(f\varphi)(V_p) \overset{\text{def}}{=} f(p) \cdot \varphi(V_p)$$

「Def. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a differentiable function. The 1-form

$$df(V_p) \overset{\text{def}}{=} V_p[f]$$

is called the differential of f.

「Lemma, For fixed $p \in \mathbb{R}^3$, $\{dx_i \mid i = 1, 2, 3\}$ is the dual basis for $T_p\mathbb{R}^3$ corresponding to $\{e_1(p), e_2(p), e_3(p)\}$.

That is, given 1-form φ,

$$\varphi = \sum_{i=1}^{3} \varphi \circ e_i \cdot dx_i .$$

Moreover, for differentiable f,

$$df = \sum_{i=1}^{3} \frac{\partial f}{\partial x_i} dx_i .$$

Proof. Let $p \in \mathbb{R}^3$ be fixed. Then $dx_i(e_j(p)) = \delta_{ij}$ Therefore, given $V = (v_1, v_2, v_3)$,

$$\varphi_p(V_p) = \varphi_p(v_1 \cdot e_1(p) + v_2 \cdot e_2(p) + v_3 \cdot e_3(p)) = \sum_{i=1}^{3} \varphi_p(e_i(p)) \cdot v_i = \sum_{i=1}^{3} \varphi_p(e_i(p)) \cdot dx_i(V_p)$$

Particular, $df(e_i(p)) = e_i(p)[f] = \frac{\partial f}{\partial x_i}$.

「Computation of forms.

Let $f(x,y,z) = (x^2-1)y + (y^2+2)z$. Then,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= 2xy\, dx + (x^2+2yz-1)\, dy + (y^2+2)\, dz.$$

So $\quad V_p[f] = df(V_p) = 2p_1p_2v_1 + (p_1^2 + 2p_2p_3 - 1)\, v_2 + (p_2^2+2)\, v_3.$

Proof.